

Project Performance Test

Date	10 February 2025
Team ID	PNT2022TMID123456
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	50

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values																																				
1.	Data Rendered	Sample of Data Rendered <table><thead><tr><th>State</th><th>Crop</th><th>Year</th><th>Production (Tonnes)</th><th>Area (Hectares)</th><th>Yield (Tonnes/ha)</th></tr></thead><tbody><tr><td>Punjab</td><td>Wheat</td><td>2022</td><td>18895000</td><td>3392000</td><td>5.57</td></tr><tr><td>Uttar Pradesh</td><td>Rice</td><td>2022</td><td>12840000</td><td>2856000</td><td>4.50</td></tr><tr><td>Madhya Pradesh</td><td>Soybean</td><td>2022</td><td>10620000</td><td>5161000</td><td>2.06</td></tr><tr><td>Maharashtra</td><td>Cotton</td><td>2022</td><td>7200000</td><td>12130000</td><td>0.59</td></tr><tr><td>Karnataka</td><td>Maize</td><td>2022</td><td>4350000</td><td>1387000</td><td>3.13</td></tr></tbody></table> - Dataset: India Crop Production (2010-2022) - Total Records: 1,25,420 - Total States/UTs: 36 - Crops Covered: 20 Major Crops - Time Period: 2010 - 2022	State	Crop	Year	Production (Tonnes)	Area (Hectares)	Yield (Tonnes/ha)	Punjab	Wheat	2022	18895000	3392000	5.57	Uttar Pradesh	Rice	2022	12840000	2856000	4.50	Madhya Pradesh	Soybean	2022	10620000	5161000	2.06	Maharashtra	Cotton	2022	7200000	12130000	0.59	Karnataka	Maize	2022	4350000	1387000	3.13
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2.	Data Preprocessing	- Missing values handled (imputed with median/mode) - Removed duplicate records - Standardized crop names and state names - Data types corrected (Year - Integer, Production - Numeric) - Created new fields: Yield (Tonnes/ha) = Production / Area - Final Cleaned Records: 1,23,876 Missing Values Summary (Before vs After) <table><thead><tr><th>Field</th><th>Before</th><th>After</th></tr></thead><tbody><tr><td>Production (Tonnes)</td><td>1,245</td><td>0</td></tr><tr><td>Area (Hectares)</td><td>1,102</td><td>0</td></tr><tr><td>State</td><td>56</td><td>0</td></tr><tr><td>Crop</td><td>78</td><td>0</td></tr></tbody></table>	Field	Before	After	Production (Tonnes)	1,245	0	Area (Hectares)	1,102	0	State	56	0	Crop	78	0																					
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3.	Utilization of Filters	- Year Filter: 2010 to 2022 (Default: All) - State Filter: All States & UTs - Crop Filter: All Crops - Season Filter: Kharif / Rabi / All - Unit Filter: Production in Tonnes, Area in Hectares, Yield in Tonnes/ha Year (All) State (All) Crop (All) Season (All)																																				
4.	Calculation fields Used	- Total Production = SUM([Production (Tonnes)]) - Total Area = SUM([Area (Hectares)]) - Average Yield = AVG([Yield (Tonnes/ha)]) - YoY Growth (%) = $\frac{SUM([Production]) - LOOKUP(SUM([Production]), -1)}{LOOKUP(SUM([Production]), -1)}$ - Top Crop by Production = RANK(SUM([Production]), 'desc') Calculated Fields (Tableau) Total Production Total Area Average Yield YoY Growth (%) Top Crop by Production																																				
5.	Dashboard design	India's Agricultural Crop Production Dashboard Total Production (Tonnes) 2.74B Total Area (Hectares) 176.35M Average Yield (Tonnes/ha) 2.92 No. of Crops 20 Production by Year Top 5 States by Production Production by Crop Production by State Yield Over Years No. of Visualizations / Graphs - 6 1. KPI Cards (4) 2. Line Chart - Production by Year 3. Bar Chart - Top 5 States by Production 4. Pie Chart - Production by Crop 5. Map - Production by State 6. Line Chart - Yield Over Years																																				
6.	Story Design	India's Agricultural Crop Production Story Overview Overall crop production scenario in India Trends Analyze production trends over the years Top Performers States and crops leading in production Insights Key insights and observations Conclusion Summary and future outlook No. of Visualizations / Graphs - 5 1. Overview Slide 2. Trends Analysis 3. Top Performers 4. Key Insights 5. Conclusion Slide																																				